

Microeconomic Theory for Experimentalists & Applied Microeconomists

Foundations, Behavior & AI-Augmented Experiments

[Instructor] · [email] · Office hours: [times/location]

[Semester Year] · [meeting times] · [room]

Course description

This course trains master’s students to *design, critique, and pilot real experiments*. It covers the microeconomic theory an experimentalist actually uses — choice under uncertainty, consumer and producer theory, information economics, behavioral economics, and causal inference — taught intuition-first: through graphs, worked numerical examples, and in-class demonstrations in which you generate the anomalies with your own choices before the theory names them.

The course’s distinguishing feature is serious integration of LLM generative agents (“Homo silicus”-style simulation) as both a pedagogical tool and a research method: every one to two weeks you run a simulation lab in which populations of AI agents play the games and face the decisions the theory describes, and you compare their behavior to the human experimental literature — and, often, to this class’s own in-class choices. The comparison is treated as a first-class intellectual question, never a gimmick: by the end you will know precisely what silicon pilots can establish (instrument debugging, pipeline validation, power analysis, framing detection, audits of deployed AI systems) and what they cannot (effect-size forecasts, welfare claims), and you will have the habits — preregistration, stability audits, honest methods paragraphs — that separate the two.

No prior programming experience is assumed. All lab code is provided complete and runs top-to-bottom in Google Colab; your modifications are confined to clearly marked **CHANGE HERE** blocks, and every notebook ships with a **DRY_RUN** mode that executes end-to-end without an API key or cost.

Learning objectives: the three exit competencies

Graduates of this course can:

1. **Read modern applied and experimental papers critically** — identify what was tested, whether the design supports the claims, and what would strengthen it.
2. **Design theory-informed experiments end to end** — subjects, treatments, incentive-compatible elicitation, identification, simulation-based power, and ethics as design substance. *This competency carries the most weight.*
3. **Use LLM generative agents to pilot, stress-test, and scale designs** before spending money on human subjects — with explicit, disciplined reasoning about where silicon subjects diverge from humans.

Format

- **Lectures** — theory plus the story of its application (classic experiments and recent papers), with live in-class choice demonstrations wherever the material allows. Expect to make paid decisions on your phone most weeks.
- **AI tutor sessions** (weekly, required) — structured conversations with a course-configured tutor AI that quizzes you, debugs your experiment designs, and adapts to your research interests. You submit the session log with a short reflection.
- **LLM simulation labs** (every 1–2 weeks) — run and modify generative-agent experiments; compare agent behavior to human benchmarks; write up findings in the course’s six-section format.
- **Experiment sessions** — in weeks 11–13 the second meeting is a hands-on session: a calibration workshop, a common-pool resource experiment, and a pricing tournament, each played for real (small) stakes.
- **Capstone** — design and simulate an original experiment; present a proposal ready for an IRB, an employer, or a PhD application. See `capstone/` in the course repository.

Assessment

Component	Weight
Weekly AI tutor logs + reflections	10%
Problem sets (10; lowest dropped)	25%
Simulation-lab write-ups	15%
Midterm: replicate + extend a published experiment	20%
Capstone (milestones 10% · paper 60% · pilot 15% · presentation 15% of this component)	30%

Problem sets mix guided calculations, “design an experiment to test this prediction” tasks (the heaviest-weighted part of every set), and silicon-pilot interpretation. *Lab write-ups* use the required six-section structure (question/benchmark → agent design → treatments → results vs. humans → divergence diagnosis → design implication). The *midterm* (due week 9) asks you to replicate and extend one published experiment using either human data from the paper or an LLM simulation, in a short paper with the same six-section discipline.

Weekly schedule

Core readings are read deeply (2–3 items per week); optional readings are for going further and for capstone mining. Full reading details, lab instructions, and deliverable checklists are in each module’s student handout in the course repository.

Wk	Module	Core readings	Due
1–2	1. Rational Choice, Preferences & Uncertainty	Kahneman & Tversky (1979); Holt & Laury (2002)	Lab 1 write-up; PS1
3	2. Consumer Theory, Demand & Welfare	Kahneman, Knetsch & Thaler (1990); Andreoni & Miller (2002)	Lab 2 write-up; PS2

Wk	Module	Core readings	Due
4	3. Producer Theory, Costs & Technology	Duflo, Kremer & Robinson (2011); Bloom et al. (2013)	Lab 3 write-up; PS3
5–6	4. Asymmetric Information, Moral Hazard & Adverse Selection	Fehr, Kirchsteiger & Riedl (1993); Lazear (2000)	Lab 4 write-up; PS4
7–8	5. Behavioral Economics: Deviations, Biases & Nudges	Kahneman & Tversky (1979) revisited; Ashraf, Karlan & Yin (2006); Madrian & Shea (2001)	Lab 5 write-up; PS5; <i>capstone question memo (wk 8)</i>
9–10	6. Causal Inference, RCTs & Structural Integration	Bertrand & Mullainathan (2004); Duflo, Glennerster & Kremer (2007)	<i>Midterm (wk 9)</i> ; Lab 6 write-up; PS6; <i>capstone theory + hypotheses (wk 10)</i>
11	7. Frontier: Development, Labor & Education · <i>calibration workshop</i>	Cunha & Heckman (2007); Miguel & Kremer (2004)	Lab 7 write-up; PS7
12	8. Frontier: Environmental, Resource & Conservation · <i>CPR experiment session</i>	Ostrom, Walker & Gardner (1992); Jayachandran et al. (2017)	Lab 8 write-up; PS8; <i>capstone design draft</i>
13	9. Frontier: Digital & Platform Markets · <i>pricing tournament</i>	Calvano et al. (2020); Blake, Nosko & Tadelis (2015)	Lab 9 write-up; PS9
14	10. AI-Econ Nexus & Capstone · <i>capstone workshop</i>	Horton (2023); Argyle et al. (2023)	Lab 10 report; PS10; <i>capstone simulation pilot + preregistration</i>
15	Capstone	—	<i>Comparison memo</i>
F	Capstone presentations	—	<i>Final paper + presentation</i>

Optional/going-further readings for every module (classics such as Allais 1953, Ellsberg 1961, Holmström 1979, Rochet & Tirole 2003; surveys such as DellaVigna 2009 and Foster & Rosenzweig 2010; and one recent LLM-agents paper per module, announced in class) are listed in the module handouts. Textbook support: Nicholson & Snyder, *Microeconomic Theory* (theory backbone, chapters assigned per module); Kagel & Roth, *Handbook of Experimental Economics* (selected chapters); Thaler & Sunstein, *Nudge* (behavioral modules, mechanism taxonomy).

Software and materials

- **Google Colab** (free tier sufficient) — all labs run here; nothing is installed locally.
- **Anthropic API key** — required for live lab runs; a modest budget (typically under \$5 per lab) suffices, and every notebook’s DRY_RUN mode works without any key. [**Instructor: state here whether course API credits are provided.**]
- **Course repository** — <https://github.com/moncap/micro-course>: labs (with Open-in-Colab links), problem sets, handouts, and capstone materials.

- **LaTeX** (Overleaf recommended) — problem-set and capstone write-ups may be submitted in LaTeX or Word; LaTeX templates are provided and its use is encouraged.
- **Expected Parrot EDSL** (optional) — for survey-style capstone simulations; plain-Python scaffolds cover everything required.
- In-class experiments run on **oTree**/Qualtrics instances operated by the instructor; you only need a phone or laptop with a browser.

AI use policy

This course *requires* AI use — and demands discipline about it.

- **Required:** the weekly tutor sessions, the simulation labs, and the capstone pilot are AI-based coursework. Problem sets flag AI-permitted parts with $\langle \text{AI} \rangle$.
- **Disclosure:** any substantive AI assistance — required or voluntary — must be disclosed, and transcripts attached where the assignment says so. Keep your logs; they are part of your grade and your protection.
- **Boundaries:** AI may draft code within the labs' **CHANGE HERE** conventions and may serve as a study partner anywhere. The analysis, interpretation, and prose of your write-ups must be your own; verbatim AI-written analysis submitted as yours is an academic-integrity violation.
- **Provenance:** simulated data must never be presented as human data, in this course or anywhere else. This is an integrity constraint, not a formality.
- **Honesty standard:** claims about what a simulation shows are graded against the course's can/cannot framework; over-claiming costs more than admitting a limitation.

Course policies

- **Collaboration:** discussing ideas on problem sets is encouraged; write-ups are individual. Lab write-ups and the capstone follow their own stated rules (capstone pairs allowed with a contribution statement).
- **Late work:** lab write-ups and problem sets lose 10% per day, capped at 50%; the lowest problem set is dropped. Capstone milestones are gating — late milestones delay feedback, not points, but the final deliverables are firm dates.
- **In-class experiments** pay real (small) stakes and are part of the course's data; participation is expected, and any student may opt out of having their (always anonymized) choices displayed by telling the instructor.
- **Accessibility:** students needing accommodations should contact [disability services office] and the instructor in week 1.
- **Academic integrity:** governed by [institutional policy]; the AI-use policy above defines the line for AI-related work.

One paragraph of advice

The students who thrive in this course are not the strongest coders or the most mathematically fluent — the code is given and the math is scaffolded. They are the ones who take the design questions seriously: who ask, of every model, “what experiment does this let me build?” and, of every simulation, “what exactly is this evidence of?” Bring your own research interests — the tutor sessions, the problem sets' design parts, and the capstone are all built to run on them.